

RESEARCH ARTICLE

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Special Section:

Rock Physics of the Upper Crust

Key Points:

- Mudstone carbonate rock was imaged with X-ray microcomputed tomography, and a calcite dissolution model was numerically implemented
- Microstructural heterogeneity caused preferred fluid pathways that localized dissolution to a subset of the total connected porosity
- The widening of narrow pores caused seemingly porosity-independent permeability evolutionary trend, which transitioned to channelization

Supporting Information:

- Supporting Information S1

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Evolution of permeability and microstructure of tight carbonates due to numerical simulation of calcite dissolution

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Abstract The current study concerns fundamental controls on fluid flow in tight carbonate rocks undergoing CO₂ injection. Tight carbonates exposed to weak carbonic acid exhibit order of magnitude changes in permeability while maintaining a nearly constant porosity with respect to the porosity of the unreacted sample. This study aims to determine—if not porosity—what are the microstructural changes that control permeability evolution in these rocks? Given the pore-scale nature of chemical reactions, we took a digital rock physics approach. Tight carbonate mudstone was imaged using X-ray microcomputed tomography. We simulated calcite dissolution using a phenomenological numerical model that stands from experimental and microstructural observations under transport-limited reaction conditions. Fluid flow was simulated using the lattice-Boltzmann method, and the pore wall was adaptively eroded at a rate determined by the local surface area and velocity magnitude, which we use in place of solvent flux. We identified preexisting, high-conductivity fluid pathways imprinted in the initial microstructure. Though these pathways comprise a subset of the total connected porosity, they accommodated 80 to 99% of the volumetric flux through the digital sample and localized dissolution. Porosity-permeability evolution exhibited two stages: selective widening of narrow pore throats that comprised preferential pathways and development and widening of channels. We quantitatively monitored attributes of the pore geometry, namely, porosity, specific surface area, tortuosity, and average hydraulic diameter, which we qualitatively linked to permeability. This study gives a pore-scale perspective on the microstructural origins of laboratory permeability-porosity trends of tight carbonates undergoing transport-limited reaction with CO₂-rich fluid.

1. Introduction

Prediction of subsurface permeability from geophysical data is crucially important to a number of applications in the geosciences, namely, hydrocarbon extraction, CO₂ sequestration, and ground contaminant research. For many rock types, including pure sandstones, permeability can be reasonably inferred from estimates of subsurface porosity using an appropriate rock physics model [e.g., Kozeny, 1927; Carman, 1937; Bourbié and Zinszner, 1985]. However, the pore geometries of tight carbonates are so complex and heterogeneous that permeability becomes nonunique, spanning nearly 4 orders of magnitude, with respect to porosity [e.g., Lucia, 1983, 1995; Vanorio *et al.*, 2015]. The lack of a functional relationship between permeability of carbonates and microstructural properties makes modeling fluid flow and interpreting subsurface hydraulic properties in carbonate reservoirs challenging.

Recent experiments [Grombacher *et al.*, 2012; Vanorio *et al.*, 2015] conducted in the transport-limited reaction regime using carbonate mudstones have shown that porosity-permeability changes of comparable magnitude can be obtained through the dissolution of calcite by injection of CO₂-rich fluid into the pore space. Interestingly, Vanorio *et al.* [2015] found that the permeability of these samples increased between 0.5 and 1 order of magnitude while the porosity remained constant within experimental error. The current study seeks to provide quantitative constraints on the physical, pore-scale parameters that control macroscopic permeability during dissolution of calcite in tight carbonate rocks.

The most commonly used rock physics model that relates permeability to pore-scale parameters is the classic semiempirical Equation Kozeny-Carmen Equation [Kozeny, 1927; Carman, 1937],

$$k = C \frac{\phi^3}{S^2 \tau^2} \quad (1)$$

which relates permeability k , to porosity ϕ , tortuosity τ , specific surface area S , and an empirical scaling constant C . For a bundle of tortuous tubes, equation (1) is exact; however, real rock microstructures exhibit complexity and spatial heterogeneity on multiple scales relevant for porous flow. Therefore, equation (1) is valid only under simple geometric transformations of idealized pore geometries or to model compaction of regularly packed granular material. Furthermore, S and ϕ are highly correlated, as they both functions on the hydraulic diameter integrated along the length of the tube [Lock *et al.*, 2002]. Determining whether a change in S or ϕ is responsible for an observed change in permeability is intractable without completely constraining the microstructure of the rock. To make matters worse, parameters S and τ cannot be directly measured on whole rocks using conventional rock physics techniques, so they are typically left as ad-hoc parameters to fit experimental data.

In order to account for the complexity of real rock pore geometries, Bourbié *et al.* [1987] developed a rock physics model to absorb the various hydraulic parameters into a single power law function of porosity:

$$k = k_0(\phi/\phi_0)^n \quad (2)$$

Here n is the so-called porosity exponent that empirically accounts for the morphology and topology of the pore geometry and k_0 is the permeability at some reference porosity ϕ_0 . Like equation (1), the porosity exponent takes on integer values for simple transformations. For example, $n = 2$ for uniform scaling of a uniform pipe lattice geometry about pore centerlines [Turcotte and Schubert, 2014]. In addition, it can be shown through pore network modeling that $n \approx 3$ for a tetrakaidekahedral grain pack having a random distribution of pore diameters [Zhu and Hirth, 2003]. This packing geometry agrees well with compaction trends of Fontainebleau sandstone [Bourbié and Zinsner, 1985; Bourbié *et al.*, 1987; Doyen, 1988; Gomez *et al.*, 2010] and sintered quartzite [Wark and Watson, 1998]. Higher values of n have also been observed for rocks near the percolation threshold [Bourbié *et al.*, 1987] and for rocks experiencing chemical alteration, which is discussed at length in reviewer paper Bernabé *et al.* [2003]. In general though, n is an all-inclusive, empirical quantity that has little to no justification outside the realm of simple pore geometries, e.g., sandstone, or idealized models of complex pore geometries.

In the last two decades, experimental inquiry has sought new ways to improve our understanding of permeability evolution in porous rocks. There are a number of experimental examples of fluid-rock interactions that exhibit changes in permeability virtually independent of porosity. Pertinent to this study, Vanorio *et al.* [2015] observed 0.5 to 1 order of magnitude increase of permeability without a measureable change of porosity—corresponding to a virtually infinite n —during injection of CO₂-rich water into stiff carbonate mudstones exhibiting negligible compaction under confining stress. Similar results were obtained by Noirié *et al.* [2004], who observed a 1 order of magnitude increase in permeability within a 0.5% increase in porosity in the early stages of CO₂ injection into carbonate packstones without confining pressure, resulting in an initial porosity exponent of $n = 150$ that decayed and then stabilized to $n = 13$. Noirié *et al.* [2004] attributed their result to the dissolution of micrite constricting hydraulic pathways. Conceptually, this makes sense, as micrite grains correspond to the phase predominantly being dissolved due to high reactive surface area [Vanorio, 2015], but a comprehensive, time-lapse analysis of the parameters describing pore morphology and local flow field is needed to lead the way to predictive models.

In addition to dissolution, precipitation reaction can also give rise to porosity-independent permeability behavior [e.g., McCune *et al.*, 1979; Xu *et al.*, 2004; Adam *et al.*, 2013]. Precipitation can be thought of as the inverse process to dissolution. Instead of removing material that may constrict pore throats, precipitation causes crystals to nucleate on pore walls, decreasing the connectivity and hydraulic radii of the pore space available to fluid flow [Bernabé *et al.*, 2003].

In this study, we seek to understand, if not porosity, what are the specific changes in the rock microstructure that control permeability evolution during reaction with CO₂-rich fluid? Rock physics experiments do not afford direct access to the pore geometry during experimentation due to the impracticality to compromise between different scales of observation, i.e., plug versus pore length scale. Therefore, in this study, we took a digital rock physics approach that involved numerical simulation of calcite dissolution in real carbonate geometries obtained using X-ray microcomputed tomography (μ -CT). The sample

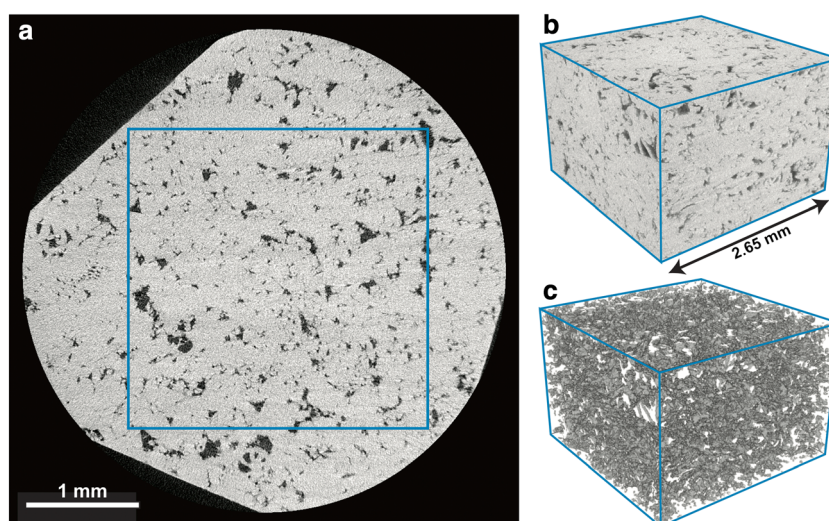


Figure 1. Visualization of tight carbonate X-ray μ -CT image. (a) Tomographic slice sampled perpendicular to the cylindrical axis of the 4.3 mm rock pillar. The gray scale value is indicative of material: black (pore) and gray (calcite). The blue box denotes the outlines the cropped subvolume. (b) Volume rendering of cropped subvolume. (c) Volume rendering of the segmented pore geometry that was used as input to the LBM and dissolution models.

used in this study is the digital representation of a carbonate mudstone used for a CO_2 -injection experiment [Grombacher *et al.*, 2012; Vanorio *et al.*, 2015]. To numerically simulate dissolution, we adaptively eroded the pore-grain interface based on local reactive surface area and local velocity magnitude, which we assumed as a proxy for solvent flux. Porosity, specific surface area, tortuosity, and hydraulic radius, were monitored over the course of the numerically simulated dissolution, and changes to those parameters are discussed in the context of their contribution to permeability evolution. Using the results and visualizations of the evolving pore geometry, we argue that porosity-independent behavior in carbonate mudstones can be induced by high dissolution rates in narrow pore throats located along high-conductivity pathways. Over time, dissolution will lead to a channelization by enhancing preexisting pathways of least hydraulic resistance, which is a different mechanism than wormholing. We also compare the results of our dissolution model with permeability-porosity trends of CO_2 injection experiments.

2. Methods

2.1. Sample Selection, Preparation, and Imaging

The rock type used in this study is a mudstone carbonate from the Peschici Formation, Italy. The initial mineralogy, texture, porosity, permeability, and pore stiffness were characterized prior to this study [Vanorio *et al.*, 2015]. The sample exhibited a tight micrite matrix with dispersed 5–200 μm diameter pores. They also possessed high pore stiffness, suggesting that the rock did not experience compaction when exposed to reactive fluid. Thus, we did not have to account for dissolution-induced compaction in this numerical study.

To fit within the field of view of the μ -CT scanner, a 4.3 mm pillar was cored from the rock. The tight carbonate pillar was imaged at a resolution of 4.42 μm per voxel using a tabletop X-ray μ -CT scanner. Figure 1a shows a visualization of a slice taken parallel to the cylindrical axis of the pillar. In the image, the gray scale value of each voxel is indicative of X-ray attenuation. The gray voxels (high gray scale value) represent calcite, while black voxels represent pores.

2.2. Data Reduction of μ -CT Data

The fluid flow code required us to crop the data to a box, so we cropped a $2.65 \times 2.65 \times 1.81 \text{ mm}^3$ subvolume (Figure 1b) from the gray scale volumetric data set, which was largest percolating subvolume that would fit within the boundaries of the digital core sample.

The gray scale volumetric data could not be used directly as input to the fluid flow software. We first segmented the gray scale subvolume, producing into a binary image consisting of voxels whose value is either 0 (calcite) or 1 (pore). To facilitate the segmentation process, we applied an anisotropic diffusion filter Anisodif3d [Kovesi, 2000], which we overloaded for computation on a graphics processing unit (GPU). The MATLAB-base implementation was based on previous image processing work [Perona and Malik, 1990; Gerig et al., 1992]. The resulting image was segmented using a global threshold (Figure 1c).

2.3. Computation of Velocity Field

Laminar flow was numerically simulated in the segmented pore geometry using the Lattice-Boltzmann method (LBM). The essence of single-phase LBM is to model fluid flow by advecting a particle collision function whose parameters are tuned to exactly mimic reproduce physics of laminar flow [Ladd, 1993]. Fluid flow is driven by a pressure boundary condition, which is implemented in the LBM model using a density perturbation function. We used an OpenMP parallelized of the code presented by Keehm [2003]. Prior to running the LBM solver, unconnected porosity was removed from the binary pore geometry to reduce computational size. To ensure that the physics in the pore throats was well resolved, the binary geometry was oversampled without interpolation by a factor of 2, so the computational resolution was 2.21 μm per voxel. The velocity field was obtained once the permeability stabilized to a steady state; i.e., permeability did not change more than 0.0001% for 40 iterations. The flux was obtained by volume averaging the component of the velocity field parallel to the imposed pressure gradient. Given the known pressure boundary condition and dimensions of the model, Darcy's law was used to determine permeability.

We benchmarked the LBM code by solving the classic flow between two infinite, parallel plates problem, which has a closed-form solution for the velocity and volumetric flow rate. In summary, we found good agreement between numerical and analytical results. For a region as few as four voxels across, the maximum error associated with the velocity and volumetric flux was 5% and 6%, respectively. Furthermore, since dissolution tends to widen pores—increasing the number of unit cells that comprise the pore throat—the error decreases as dissolution progresses.

2.4. Dissolution Model

We numerically simulated calcite dissolution using a phenomenological approach. Rather than solving the advection-diffusion equation directly [e.g., Jiang and Tsuji, 2014; Pereira Nunes et al., 2016a, 2016b; Soulaire et al., 2016], our model assumes the local velocity field as a proxy for solvent flux. By doing so, we implicitly ignore diffusion, which we argue is a valid simplification for a transport-limited, high-flow dissolution regime where solvents are advected through the pore geometry faster than they can change the chemistry of the fluid due to reaction with the pore wall. Though our model is not first-principles-based, it builds upon basic physicochemical principles that underpin transport-limited reaction behavior, namely, (1) the reaction in nonpercolating pathways is fully buffered and (2) the reaction is most rapid in regions where solvent flux is high.

Our approach is equivalent to solving the zeroth-order reaction rate equation at each boundary grain voxel i ,

$$\frac{dm_i}{dt} = -n_i \frac{v_i}{v_{\max}} R_0 \quad (3)$$

where m_i is mass and R_0 is the maximum reaction rate, which we assumed was $R_0 = 2.57 \times 10^{-4}$ mol/s, an empirical value given by Plummer et al. [1978] for the forward reaction of calcite with weak carbonic acid. We imposed dissolution at high-flux regions by scaling R_0 by the ratio $v_i/v_{\max}$, where v_i is the mean velocity magnitude calculated in a five-voxel neighborhood of i and $v_{\max}$ is the maximum velocity in the pore volume. The dependence of reaction rate on surface area was accounted for by parameter n_i , which scales the reaction rate at each voxel by the number of voxels faces exposed to the virtual reactive fluid.

In order to enable a continuous erosion of the pore wall, we first converted the binary pore geometry into a porosity field. As dissolution proceeded, voxel porosities increased. In Figures 2a and 2b, we show the local surface area and relative velocity magnitude, respectively, and in Figure 2c, we show the local porosity at boundary voxels. Once the porosity reached a value of 1, grain voxels were converted to pore voxels. If the amount of material that was dissolved during a single iteration was more than the volume of the

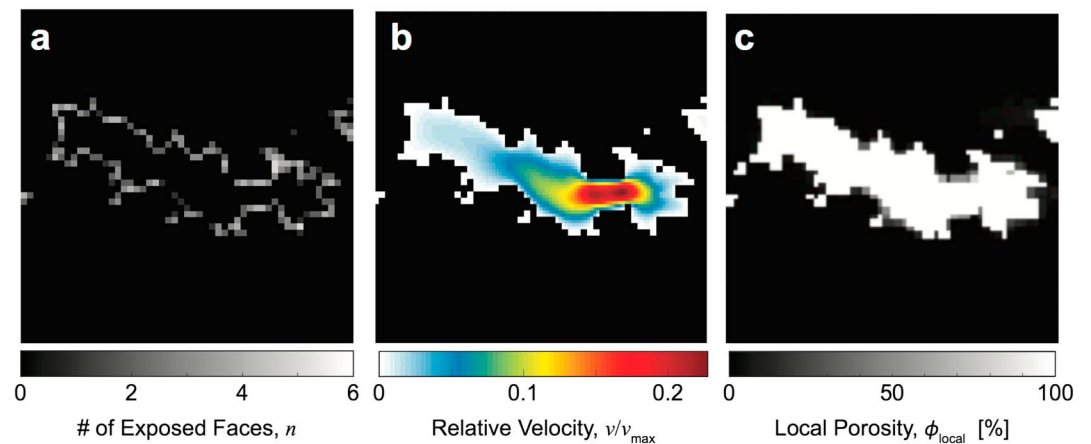


Figure 2. Visualization of the dissolution criteria, given by equation (3), around a single example pore. (a) Number of pore-wall voxel faces exposed to the virtual fluid. (b) Relative velocity magnitude. (c) Local porosity field on the pore-wall after 20 dissolution iterations (0.21 h). Note that lower porosity indicates higher local dissolution rate.

voxel, the remainder was distributed among voxels within the neighborhood defined by a 26-voxel cubic structure element.

The dissolution time step (Δt) was chosen to ensure numerical stability; no more than one third of a voxel could dissolve at a single time step. To prevent the dissolved pore geometry from wandering from the original pore geometry, the model computed the Euclidean distance map [Kimmel *et al.*, 1996] after every dissolution iteration. If the maximum wander exceeded 1.5 times the voxel length, dissolution ceased and the resulting geometry was used as input to the LBM code to compute the velocity field. Each time the LBM code was initialized, the pore geometry was reevaluated for connectivity, allowing previously unconnected porosity to become connected if it intersected with the updated pore geometry.

2.5. Characterization of Hydraulic Parameters

The geometric parameters of interest in this study were connected porosity, tortuosity, specific surface area, and hydraulic diameter. The connected porosity was determined simply by the number of voxels belonging to the connected pore binary volume divided by the total number of voxels in that volume.

Tortuosity was calculated directly from the velocity field outputted from LBM code with the following equation [Duda *et al.*, 2011]:

$$\tau = \frac{\langle u_{\text{mag}} \rangle}{\langle u_{\parallel} \rangle} \quad (4)$$

where u_{mag} is the local velocity magnitude, $u_{\parallel}$ is the local velocity parallel to the flow axis, and the notation $\langle \dots \rangle$ indicates a volume-average over the pore volume. Equation (4) provides an easy way to estimate tortuosity, and as Duda *et al.* [2011] showed, equation (4) is mathematically equivalent to computing tortuosity from average length of streamlines if those streamlines were weighted according to the volume flux integrated along each streamline.

The specific surface area was computed by summing the local surface area of the pore-solid interface and then dividing by the volume of the whole digital rock. The local surface area was obtained by convolving the inverse of the binary pore geometry with a 26-voxel structure element. Then a binary mask composed of pore-wall voxels was applied.

3. Results

3.1. Velocity Spatial Distribution

In Figure 3, we show volume renderings of the velocity evolution for different times during the dissolution simulation. In the figure, the velocity is normalized by the maximum velocity in the pore geometry. Even in

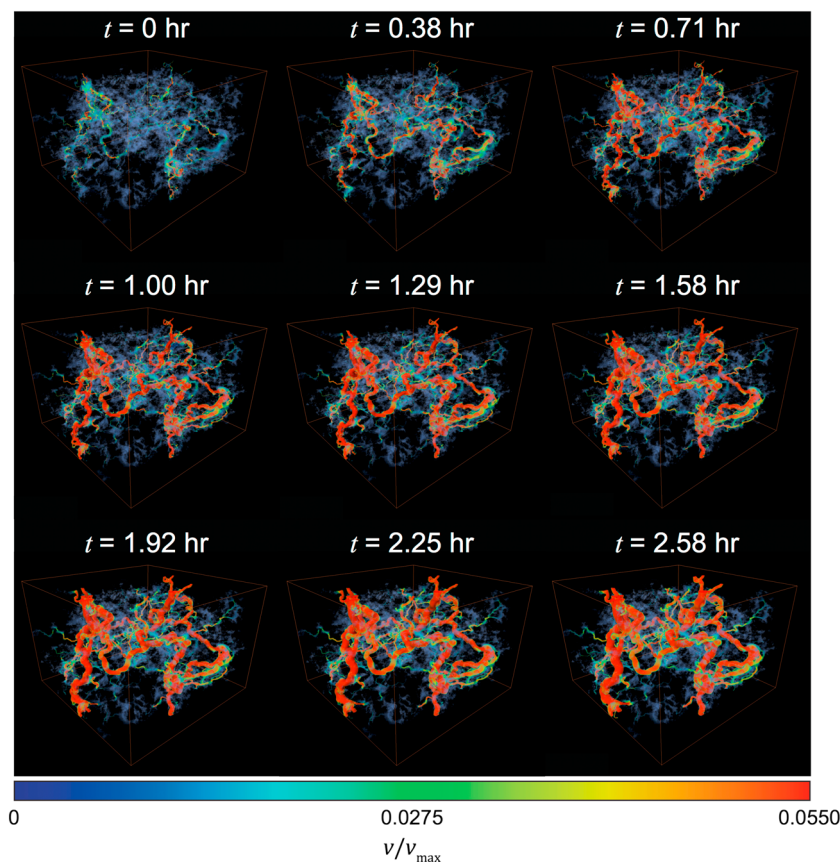


Figure 3. Evolution of velocity during dissolution simulation. Zero-velocity regions are faded to enhance visualization of the proportion of the pore space that conducts fluid flow.

the initial pore geometry ($t = 0$ h), fluid flow is highly channelized, with only a subset of the total porosity along at least three pathways having a relative fluid velocity greater than 0.01. Over the course of dissolution, the relative velocity magnitude increases only within pores that reside along these high-velocity pathways, while flow in the remaining pores is comparatively stagnant.

3.2. Evolution of Pore Geometry

We monitored the pore geometry during simulated dissolution. In Figure 4, we plot the evolution of the pore geometry of the digital carbonate rock due to the numerically induced calcite dissolution. The initial, unreacted connected pore geometry is displayed at $t = 0$ h (gray) and faded for all subsequent times to enhance the visualization of the porosity created by or made available to flow via dissolution (red).

The time-lapse visualization of the pore geometry points to two stages of evolution. During Stage 1 ($t = 0$ – 0.71 h), dissolution was sparse, limited to patches dispersed throughout the rock volume. Yet those patches appear to be spatially correlated with the high-velocity regions in Figure 3. During Stage 2 ($t = 0.71$ – 2.58 h), those isolated regions grew along the same hydraulic pathways, until they merged, then becoming channels. The newly formed porosity was composed of voxels that had been converted from calcite, rather than by preexisting, unconnected pores that became available to flow through dissolution. Though we allow unconnected porosity to become connected during dissolution, connectivity does not appear to increase significantly throughout dissolution. Instead, dissolution widens preexisting, connected pores along high-velocity pathways.

For a pore-scale view of the evolving pore morphology, we also plot in Figure 5a contours of the connected pore geometry on a two-dimensional cross-section sampled orthogonal to the flow direction. Each contour represents a time snapshot of the binary connected pore geometry. Two example pores are shown in detail in Figures 5b and 5c.

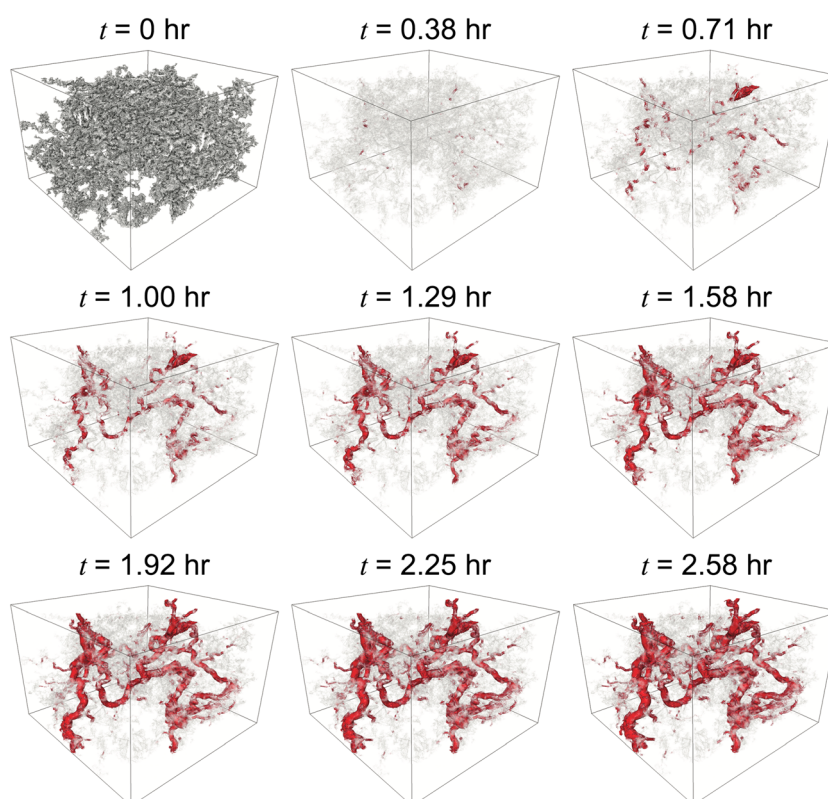


Figure 4. Volume renderings depicting the evolution of the connected pore geometry during dissolution simulation. Voxels created or made accessible to flow through dissolution are colored red. The initial pore geometry is shown in gray at $t = 0$ h and is faded for all subsequent times ease visualization of the created pore space.

During Stage 1, dissolution was characterized by smoothing of the pore wall; dissolution only removed pores that having the highest local surface area. With the exception of those high-surface area pores, the pore diameter measured on the cross section remained approximately constant at ~ 60 μm . By $t = 0.71$ h, dissolution occurred along the entire perimeter of the pore wall. Each time step advanced the position of the pore wall

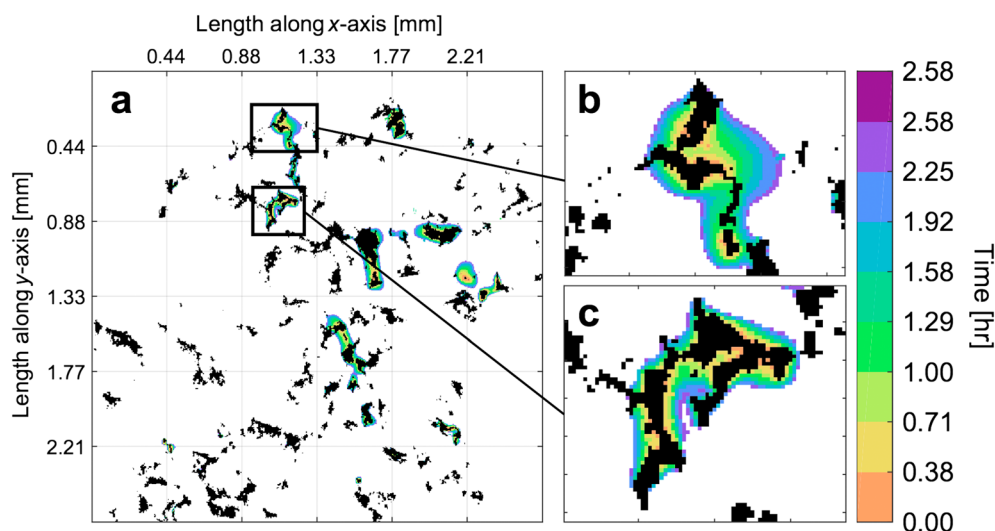


Figure 5. Dissolution overprint. (a) Slice of binary pore geometry overlaid by contours of the dissolution history. Slice is oriented orthogonal to the flow axis. (b and c) Close-ups of two select regions in Figure 5a that depict the evolving pore geometry.

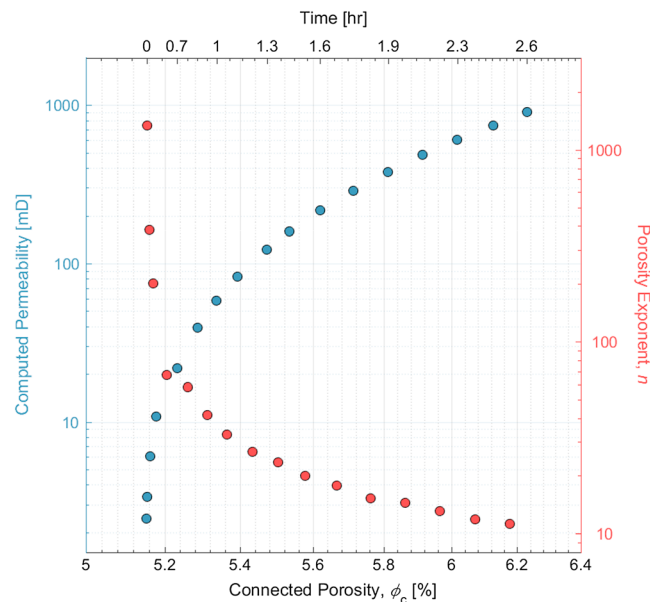


Figure 6. Permeability (blue) and porosity exponent (red) plotted as a function of connected porosity (bottom x axis) and time (top x axis). The x and y axes are plotted in log-scale so that the porosity exponent (n) is the instantaneous slope of permeability-connected porosity curve.

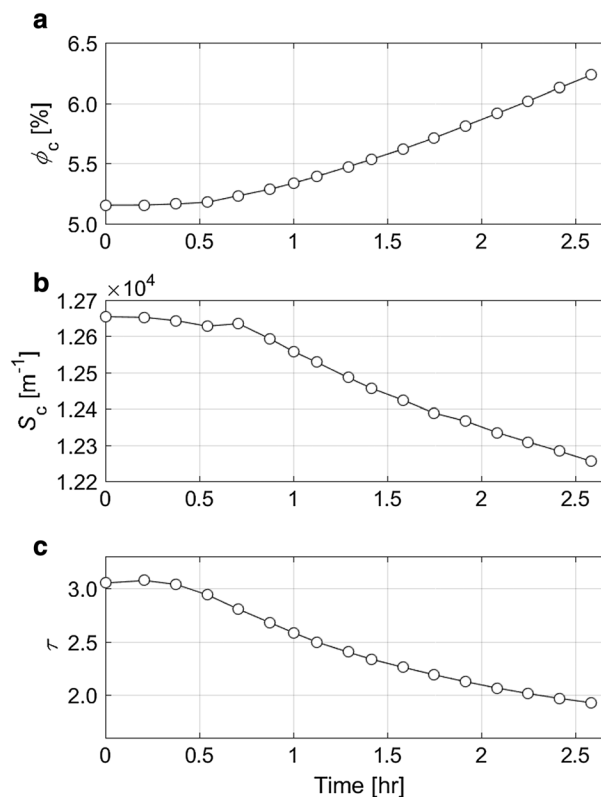


Figure 7. Evolution of connected pore geometry hydraulic attributes. (a) Porosity of the connected pore geometry, ϕ_c , versus time. (b) Specific surface area of connected pore geometry, S_c , versus time. (c) Tortuosity, τ , computed directly from the velocity field using equation (4). results of calcite dissolution simulation.

further into the calcite matrix and tended to round the pore. Although, the pores retained by of their original shape and had not advanced so far by $t = 2.58$ h to have become tubular. By the end of the dissolution simulation, at $t = 2.58$, the diameter of pores were approximated $100 \mu\text{m}$.

3.3. Permeability and Attributes of the Pore Geometry

We plot the permeability versus connected porosity trend in Figure 6 (blue circles). Permeability increased from 2 mD to 906 mD from the initial pore geometry to the final, dissolved pore geometry at $t = 2.58$ h. The increase of connected porosity due to dissolution over the same time interval was small by comparison, only $\sim 1\%$ of the initial value at $t = 0$ h.

We also give the porosity exponent n (red circles), which quantifies the rate of change of $\log_{10}(k)$ with respect to $\log_{10}(\phi_c)$. The evolution of n indicates two stages of the pore geometry evolution. During Stage 1 ($t = 0-0.71$ h), n decreased from 1334 to 67, the largest increase of permeability with respect to connected porosity during the simulation. During Stage 2 ($t = 0.71-2.58$ h), n decreased from 58 to 6. There is some evidence of a transitional stage between $t = 0.71-1.29$ h where n behaves nonlinearly in log-log space. Afterward, n decreases more or less linearly until the end of the dissolution simulation. It is unclear whether the porosity exponent would have converged on a constant value if the simulated dissolution had continued, which would have indicated uniform dissolution along the length of the channels.

In addition to permeability, we also track attributes of the pore geometry relevant to fluid flow. Our choice of parameters, namely, connected porosity ϕ_c (Figure 7a), specific surface area S_c (Figure 7b), and tortuosity τ (Figure 7c), was motivated by Kozeny-Carman formalism

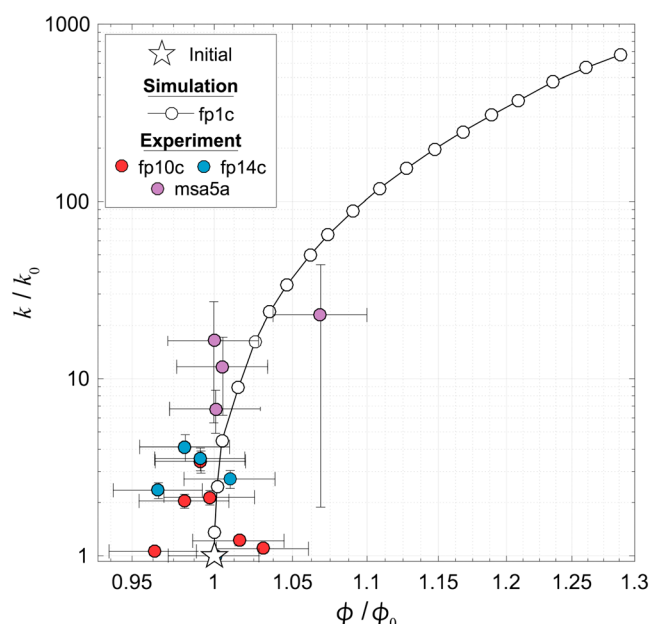


Figure 8. Permeability-porosity comparison of dissolution simulation results and CO₂-injection trends conducted on natural carbonate mudstones [Vanorio *et al.*, 2015]. Initial porosities (ϕ) and permeabilities (k) of both experimental and digital samples were different, so porosities and permeabilities were normalized by the porosity (ϕ_0) and permeability (k_0) of the initial, unreacted sample. The star marker represents the porosity and permeability of the initial normalized porosity and permeability. The colored markers represent experimental results. The error bars on experimental data represent 2σ uncertainty of the normalized porosities and permeabilities, which was calculated by propagating the error in the measured porosity ($\sigma_\phi = 1\%$) and permeability ($\sigma_k = 2\%$). The empty markers represent the simulation results from this study.

ized porosity (ϕ/ϕ_0) and permeability (k/k_0) values. The error bars on experimental values in Figure 8 represent the two-sigma uncertainty values. The normalized porosity and permeability values for the simulations results are also plotted.

We want to emphasize here that direct comparison of rock physics data is not the primary goal of this research as it poses two difficulties. First, the volume use for digital rock physics simulation in this study is order 1000 times smaller than that use for experiments. We do not presume that our choice of sub-volume is representative of the samples. Given the high degree of heterogeneity present in carbonate samples, there is no guarantee that the same microstructures are swept by the reactive fluid in experiments and simulations. Second, compaction—which we explicitly ignore in our simulations—may hamper permeability evolution by closing compliant pores during the dissolution process [Vanorio *et al.*, 2011]. Though carbonate mudstones are relatively stiff, compared with carbonate packstones, they do experience some ($\sim 0.2\%$) compaction subject to confining pressure during CO₂ injection experiments [Vanorio, 2015].

Nevertheless, our simulation does seem to agree with experiments in terms of porosity-permeability relationship during Stage 1. The experimental data, with the exception of three data points, fall within uncertainty of simulation results, so we infer that porosity-independent permeability behavior of experiments is primarily due to dissolution; precipitations of CaCO₃ in the pore space and compaction are likely secondary processes. Furthermore, the fact that experimental data overlap only Stage 1 of our simulation results suggests that channels did not form in the experimental samples, which is corroborated by microstructural analysis of the injected samples [Vanorio *et al.*, 2015]. However, we cannot say at this point that the absence of channel formation in injection experiments is due to insufficient duration of the experiment or to hampering by compaction.

(equation (1)). Despite the permeability increase during Stage 1, ϕ_c , S_c , and τ were constant. During Stage 2, porosity increased while specific surface area and tortuosity decreased, consistent with equation (2).

4. Discussion

4.1. Comparison With Experimental Data

Here we compare our simulation results with CO₂ injection experiments. We plot in Figure 8 our simulation and experimental results for three carbonate mudstones [Vanorio *et al.*, 2015]. For experimental data, porosity was measured using He porosimetry. Initial porosities and permeabilities of the three samples ranged 4.3–10.6% and 8–222 mD, respectively. To compare our simulation results with experimental data, we normalized the experimental and simulated porosity and permeability data by those of their respective unreacted samples. Uncertainty of the measured porosity and permeability values for the injection experiments were 1% and 2%, respectively, which we propagated to the normal-

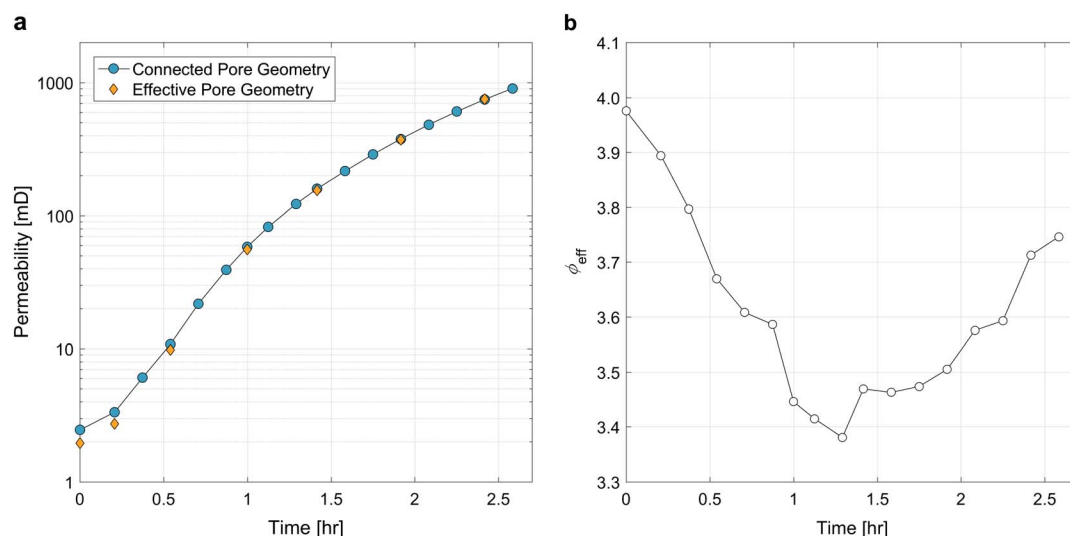


Figure 9. Evolution of the attributes of the effective pore geometry. (a) Permeability of connected and effective pore geometry versus time. (b) Effective porosity, ϕ_{eff} , versus time.

4.2. From Preferred Pathways to Channels

The spatial distribution of velocity in the carbonate mudstone pore geometry is spatially heterogeneous. As is evident by the velocity magnitude visualization at $t = 0$ h in Figure 3, flow is highly channelized, with a fraction of porosity accommodating a disproportionate share of the fluid flux. These so-called preferential fluid pathways have also been observed in modeling studies concerning fluid flow in random pore media [Martys and Garboczi, 1992; David, 1993; Yang et al., 1993; Pyrak-Nolte and Montemagno, 1994] as well as in real rock geometries [Liu et al., 2005; Karpyn et al., 2007; Smith et al., 2013; Miller et al., 2015; Menke et al., 2016]. They are best interpreted as pathways of least hydraulic resistance that result from heterogeneity imposed by the complex depositional and diagenetic history of the rock.

The coupling between velocity and dissolution rate constitutes a positive feedback that localizes dissolution and high flow rates to preferential fluid pathways. Thus, the heterogeneity of the initial rock microstructure acts as an initial condition to the dissolution reaction and evolution of the pore geometry. At the onset of dissolution, pores having the highest velocity surface-to-volume ratio residing along these preferential fluid pathways dissolve first. Throughout Stage 1, these regions grow until dissolution occurs along the entire length of the preferential fluid pathways, forming channels. By Stage 2, the permeability of the rock is controlled these channels, which spatially correlate with the locations of the preferential pathways present in the initial, unreacted pore geometry.

4.3. Geometric Attributes and Porosity-Permeability Behavior

Despite the 1 order of magnitude increase of permeability during Stage 1, Kozeny-Carman parameters, namely, connected porosity, specific surface area, and tortuosity, in Figure 7 are constant. One interpretation of this seemingly unintuitive result is that a significant proportion of pores containing comparatively stagnant flow, yet contribute to the total porosity and surface area. These pores can be thought of as a background signal, and the porosity created through dissolution is comparatively small. In our digital pore geometry, stagnant pores are those that do not reside along preferred fluid pathways. In order to isolate late the changes to permeability to those pores that actually accommodate fluid flow, we make a distinction between effective and ineffective pores. To this end, we labeled individual pores by applying a morphological watershed algorithm on the connected pore geometries and used the velocity field to calculate the flux into each pore. We then removed ineffective pores, defined as pores whose flux was less than a threshold value, which we denote as a tunable parameter p . We find that a value of $p = 0.05$, or 5% of the average pore flux, satisfactorily reproduces the permeability trend within 0.1–20% of the connected pore geometry permeability (Figure 9). More details on the methodology for partitioning pores into effective and ineffective porosity can be found in

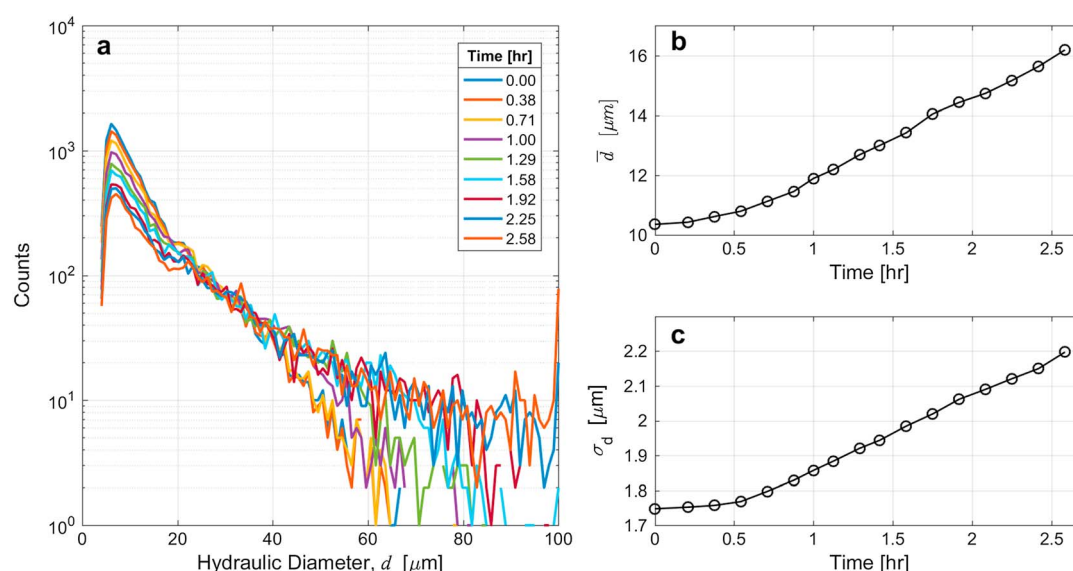


Figure 10. Evolution of the hydraulic diameter as measured using a skeletonization of the effective pore geometry. (a) Hydraulic diameter histogram of the effective pore geometry. (b) Time evolution of the geometric mean hydraulic diameter. (c) Time evolution of the geometric standard deviation hydraulic diameter.

the Online Supplement [Beucher and Meyer, 1992; Meyer, 1994; Kimmel et al., 1996; Bernabé et al., 2003; Porikli and Tuzel, 2006; Klasing et al., 2009; Holzer et al., 2012].

Interestingly, dissolution gives rise to some unintuitive porosity behavior, given in Figure 8b. Conventional thinking, driven by the Kozeny-Carman formalism, is that permeability scales with the cube of porosity. However, the effective porosity first exhibits a decrease from $\phi_e = 4.0$ –3.4% during the period $t = 0$ –1.29 h, followed by an increase from $t = 1.29$ –2.58 h. We argue that this behavior is a result of localized dissolution at preferential fluid pathways. That is, as dissolution widens preferential fluid pathways and decreases the tortuosity of the pathways, they accommodate a larger fluid flux, so much that less conductive pores are dropped from the fraction that comprises the effective pore geometry. The dropped porosity outweighs the porosity produced through dissolution of narrow pores, so the total effective porosity decreases during Stage 1. As the narrow pores then continue to widen, the velocity distribution along each preferential path becomes more uniform, which, in turn, induces dissolution over a larger surface area. By $t = 1.0$ h, dissolution occurs over the entire length of every percolation pathway, resulting in a net porosity gain for all subsequent times.

4.4. Hydraulic Diameter

Despite their widespread use in rock physics, the singularly interesting porosity behavior suggests that it is not a good indicator of permeability evolution during dissolution. More fundamentally than porosity or specific surface area, permeability depends on the mean hydraulic diameter integrated along flow paths [Lock et al., 2002]. In order to ascertain the time evolution of the hydraulic diameter distribution, we skeletonized the effective pore geometry using a medial-axis thinning algorithm [Lee et al., 1994; Kerschnitzki et al., 2013] and then retrieved the hydraulic diameter at each point along the skeleton by interpolating the Euclidean distance map. Because the distance map given the distance to the nearest calcite voxel for every pore voxel, the hydraulic diameter is the circumscribed in the cross section of the pore. More information about obtaining the hydraulic diameter through skeletonization is listed in the supporting information [Lee et al., 1994; Zhu et al., 2011; Kerschnitzki et al., 2013; Miller et al., 2014].

We plot the hydraulic diameter distributions in Figure 10a along with the time evolution of the geometric mean and standard deviation (Figures 10b and 10c). Over the course of dissolution, there is an inversion of the pore diameter about ~ 25 μm , which we interpret as conversion narrow pore throats into wider pores, and is consistent with experimental observations [Shulakova et al., 2016]. Despite the stagnancy of the porosity and specific surface area of the connected pore geometry (Figures 7a and 7b) and unintuitive

effective porosity behavior (Figure 9b), the hydraulic diameter best mimics—at least qualitatively—the permeability evolution. Furthermore, the mean hydraulic diameter in Stage 1 exhibits a slower rate of change than Stage 2, which temporally correlates with the transition from patchy dissolution to channelization illustrated in Figure 4. Therefore, we propose that the evolution of the permeability and pore morphology can be conceptualized as first selective widening of high-velocity, narrow pore throats, followed by widening along the entire length of newly formed channels.

As a conceptual example, consider flow through a series-connected pore configuration. Though the volumetric flux in each of the pores is the same due to mass conservation, velocity and the dissolution rate are highest in the narrowest pores, causing those pores to widen faster than other pores in the network. As the narrow pores widen, the velocity magnitude distribution within each preferred fluid pathway—not on the whole pore geometry—becomes progressively homogeneous, which in turn induces dissolution over a larger surface area, eventually inducing dilation over the length of the entire channel.

The two-stage porosity-permeability trend supports this conceptual dissolution behavior model. For the series pore configuration given above, permeability is proportional to the harmonic mean hydraulic diameter and thus is strongly weighted by the narrowest pore in the configuration. Also, the narrowest pores also contribute the least to the total pore volume, which are largely outweighed in the calculation of the geometric mean, explaining both the minute porosity increase and slow rate of change of the mean hydraulic diameter during Stage 1. Therefore, widening of the narrowest pores maximally impacts permeability while minimally impacting porosity, a possible explanation for porosity-independent permeability behavior. During Stage 2, all pores along the preferred fluid pathways, which make up a nonnegligible portion of the porosity and number of pores, widen.

Though a thorough analysis is outside the scope of this study, given the widespread use of remote sensing techniques to detect changes in the subsurface, we briefly discuss the elastic properties of our digital rock. Vanorio *et al.* [2015] and Vanorio [2015] measured the primary wave velocity (V_p) for carbonate mudstones as a function of injected pore volumes and confining pressure. They showed that the V_p decreases during dissolution by ~ 0.1 km/s and is insensitive to within measurement error of the measured porosity. Since changes in V_p are sensitive to changes in the bulk porosity, we do not expect the small modifications to pore throats observed in our numerical simulations to measurably affect the wave speeds.

4.5. Channelization Versus Wormhole Formation

Though the morphology of channels in this study bare a strong resemblance to so-called wormholes [e.g., Fredd and Fogler, 1999; McDuff *et al.*, 2010; Smith *et al.*, 2013, 2014], we emphasize here that they form via different processes and under different flow conditions. Given the same initial chemistry of the reactive fluid, wormhole formation depends both on the flow rate and the initial microstructure of the rock. In a low-flow regime, solvents react with calcite faster than they can be advected through the pore space. As a result, the reaction front literally burrows a channel into the rock at a rate and direction that depends on the distribution of lithology, porosity, and reactive surface area within the sample [Smith *et al.*, 2014]. In contrast, in a high-flow regime, solvents are advected into the pore space faster than they can dropout by reacting with the CaCO_3 pore wall. As was pointed out in a recent study [Menke *et al.*, 2016] and we confirm with simulations, the channels formed in a high-flow regime is not due to wormholing but is related to preexisting heterogeneity of the pore geometry.

The distinction between wormholing and channelization is subtle but important in the context of permeability-porosity evolution. Permeability of a rock is controlled by the smallest pore throat along percolation pathways. However, wormholing, prior to breakthrough, modifies only the pores at the inlet of the sample. As a result, the porosity exponent should be nearly zero during wormholing prior to breakthrough. In contrast, the modifications to the pore space we observe in our simulations, especially at the onset of dissolution, are so small yet impactful that permeability increases over an order of magnitude without a measureable change of porosity.

Wormholing and channelization may also have different impacts on the mechanical properties of rocks. The pore spaces produced by wormholing in a low-flow regime are so large that they may cause the rock to compact. Conversely, injection experiments [Vanorio *et al.*, 2015] show that the rock can remain stiff even during some channelized dissolution. The contrasting elastic behavior between wormholing in low-flow regime and

channelization in high-flow regime has importance implications for remotely monitoring rock physics properties in reservoir during enhanced hydrocarbon recovery or CO₂ sequestration methods.

5. Conclusion

Numerical simulations of calcite dissolution were conducted on digital carbonate mudstones obtained from μ -CT imaging to elucidate changes in microstructure that could explain the porosity-independent permeability behavior observed in CO₂ injection experiments of the same rock type. The goal of the model was to produce a nonuniform transformation of the pore geometry similar to dissolution in a transport-limited regime, compute permeability, and relate changes in permeability to specific attributes of the dynamic pore geometry. As a first approach, we developed a simple, phenomenological dissolution model that coupled the dissolution rate of the pore wall to the neighborhood velocity field obtained through LBM. The digital pore geometry was evolved through time by iterating between simulation of dissolution and fluid flow. At each LBM computation, we calculated permeability from the velocity field, which we linked to changing attributes of the pore geometry.

We found that preexisting heterogeneity in the rock microstructure gave rise to preferred fluid pathways, which accommodated a disproportionate share of the flow rate. Since dissolution rate was coupled to the local velocity field in our model, dissolution only occurred along these pathways. The permeability-porosity evolution of the digital pore geometry appeared to have two stages. Stage 1 was characterized by the dissolution of the narrowest pore throats of these preferential pathways. Though small compared to the total pore volume, widening of narrow pores at a faster rate than others on the same fluid pathways induced an order of magnitude increase of permeability while maintaining a virtually constant porosity. During Stage 2, dissolution occurred along the entire length of the percolation pathways, resulting in a much larger increase of porosity with each subsequent dissolution iteration. Stage 1 simulation results compare favorably with CO₂ injection experiments conducted on tight carbonate mudstones.

We also found that the pore geometry could be partitioned into two porosity classes: effective and ineffective porosities. We found that the same permeability—within a maximum error of 20%—could be obtained by removing pores with comparatively stagnant flow rates. Despite the increase of permeability through the entire dissolution simulation, we found that the effective porosity exhibited unintuitive behavior, sometimes decreasing while permeability increased. Therefore, we concluded that porosity is a poor indicator of permeability for tight carbonates undergoing dissolution from CO₂ injection. Instead, we found that the hydraulic diameter, which increases throughout the dissolution simulation, is a better indicator of the permeability behavior.

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